

Will Ferguson

LinkedIn - Github - william.ferguson83@gmail.com

Computational scientist with 10+ years building simulation, modelling, and data systems across nuclear, defence, financial, and identity sectors. Experienced in taking complex physical and mathematical problems from research to production code, with a track record across safety-critical, regulated, and data-intensive domains. Skills include software engineering, data analysis, mathematical modelling, statistics, and machine learning. SC cleared.

Languages: Python, Fortran, C++, R, Golang, SQL

Tools: Linux, Git, Docker, GCP, CI/CD (Github Actions), Pipelines (Dagster), HPC (MPI, OpenMP)

Sectors: Nuclear, Defence, Identity Tech, Proptec, Finance

EDUCATION

- **Imperial College London / CERN:** PhD, High Energy Physics 2014
- **University of Birmingham:** MSc, Physics and Technology of Nuclear Reactors 2008
- **Imperial College London:** MSc, Advanced Mechanical Engineering 2006
- **Imperial College London:** BSc, Physics 2005

PROFESSIONAL EXPERIENCE

- **Skyral (formerly Improbable Defence)** Jan 2024 - Apr 2026
Software Engineer (Modelling)
 - Dynamic route costing using the Valhalla open source routing engine
 - Development and validation of agent-based models for urban digital twins for populations of 5M+
 - Modelling of electrical power network for vulnerability analysis
 - Development of road network and synthetic population pipelines with Dagster
- **Jacobs** Jan 2021 - Jan 2024
Nuclear Reactor Physicist
 - Development and support of a widely used Monte Carlo neutron transport and reactor physics code (MONK)
 - Implementation of a set of novel depletion solvers using numerical methods for increased speed and accuracy
 - Development of a new nuclear burnup data model, and corresponding new HDF5 interface file
 - Multiphysics coupling of neutronic and thermal-hydraulic codes
 - Delivery of customer training courses
 - Mentoring of interns
- **Yoti** Sep 2014 - Sep 2019
Data Scientist
 - PoC of identity ranking algorithm in (PageRank) in C++
 - Identity attribute confidence graph-based models (eg Bayesian networks, electrical networks)
 - Implementation and testing of Stellar Consensus Protocol (blockchain) in Golang
 - Prediction market PoC in R (based on logarithmic market scoring rule)
- **Homeseller** Sep 2014 - Mar 2018
Data Scientist
 - Design and implementation of proprietary home valuation algorithms to make sale and rental valuations for UK homes, using a repeat sales regression model (Python, R). Accuracy $\pm 8\%$ (comparable with Zoopla estimates).
- **Gamesys** Apr 2014 - Sep 2014
Financial Modeller (contract)
 - Foreign exchange volatility modelling for a high frequency binary options trading platform
- **UKAEA Fusion, Culham** Jun 2008 - Sep 2008
Masters Student
 - Development of a method for applying finite element neutron transport code to a CAD model of the ITER fusion reactor, for comparison with Monte Carlo modelling